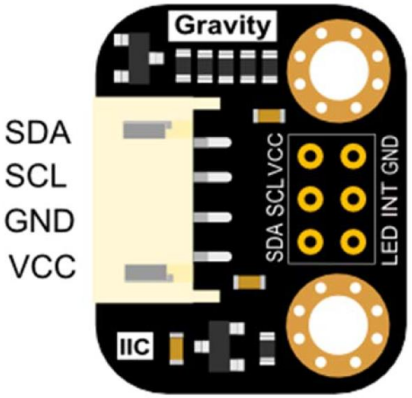
Embedded projects 2
Weekly exercises 10

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RGB – Color sensor

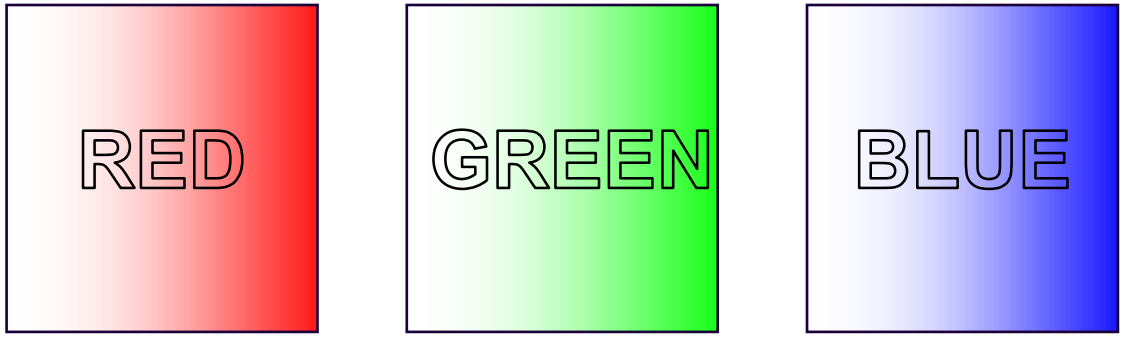


- Study the operation of the TCS34725 - **I2C** color sensor
  - The data sheet can be found from Moodle or online ↓
  - Connect the sensor using I2C, not the analog pins like in one of the examples!

- External information:
  - https://www.dfrobot.com/product-1546.html
  - https://wiki.dfrobot.com/TCS34725_I2C_Color_Sensor_For_Arduino_SKU__SEN0212

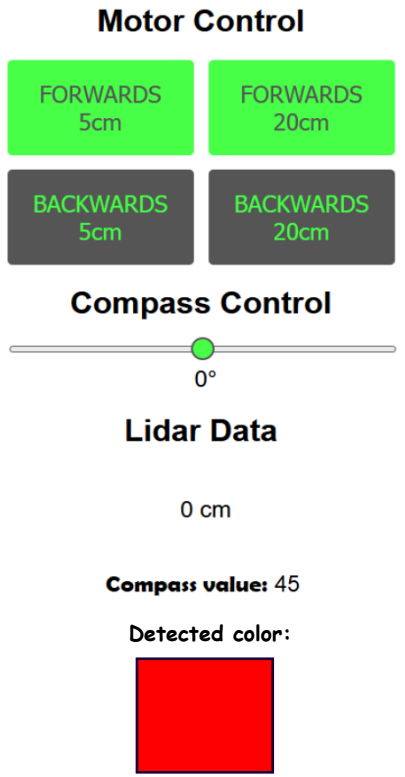
Exercise 1 – Test the sensor

- Write **a simple test code**, test with different colors, and verify that your system works
  - Notice how well sensor works and from how far? (**Crucial**)
  - Consider where you should place the sensor, if we later on want to detect the color under the car (sensor will be used to read the color of “road lines”)
- There is example code in the documentation.



Exercise 2 – Send RGB data to ESP

- Similar to sending the lidar data, send the RGB-data from sensor to your ESP/web page
- On your webpage:
  - Either, show the color with the biggest value in a separate box, for example:
    - If red = 100, blue = 150, green = 70, your color box should be blue
    - If red = 200, blue = 150, green = 70, your color box should be red
  - Or, show the actual detected color using all of the values
    - For example (255, 0, 255) would be
- Note the next exercise also



Exercise 3 – Improve your web site

- Make your website look *unique* and fabulous ☺
  - You can change basically everything, as long as all of the functionality remains (i.e. ability to drive the car and show its data on the page)
- Points are given based on usability and changes in:
  - layout
  - color scheme
  - element styling (box & slider appearance etc.)